$$f(x_1, x_2 | w_1, w_2, w_0) = \underbrace{w_0 + w_1 x_1 + w_2 x_2}_{w^T \cdot x + w_0}$$

$$\frac{\partial f(x_1, x_2)}{\partial x_1} = w_1$$

$$\frac{\partial f(x_1, x_2)}{\partial x_2} = w_2$$

where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \mathbf{w}$$

$$\frac{\partial w^T \cdot x}{\partial x} = w$$

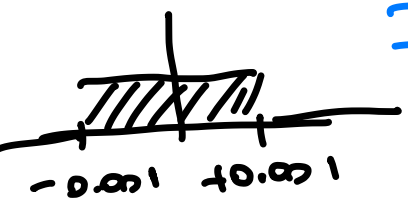
$$\frac{\partial w^T \cdot x}{\partial w} = x$$

$$\frac{\partial (w^T x + w_0)}{\partial w_0} = 1$$

Linear Discrimination for Binary Classification

STEP#1: Initialize $\{w, w_0\}$ randomly and decide η.

Initializing them to very small values; for example



Uniform$(-0.001, +0.001)$

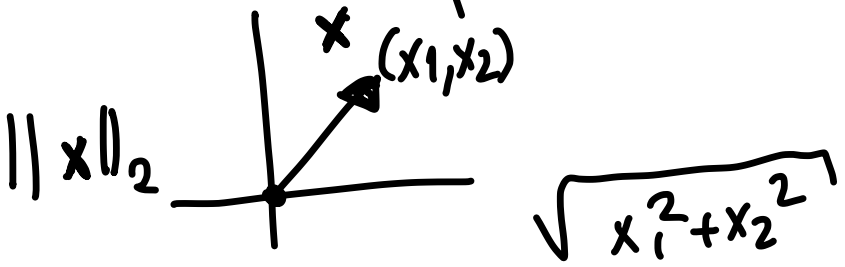
STEP# 2: Calculate Δw and Δw_0.

STEP#3: Update w and w_0.

$$w^{(t+1)} = w^{(t)} + \Delta w \longrightarrow -\eta \cdot \frac{\partial Error}{\partial w}$$

$$w_0^{(t+1)} = w_0^{(t)} + \Delta w_0 \longrightarrow -\eta \frac{\partial Error}{\partial w_0}$$

STEP#4: Go to STEP#2 if there is a change in the parameters. [i.e., $\|\Delta w\|_2 \neq 0$, $|\Delta w_0| \neq 0$]



Linear Discrimination (Multiple Classes) ($K > 2$)

$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$ $\quad x_i \in \mathbb{R}^D$ $\quad y_i \in \{1, 2, \dots, K\}$ → reference class

$$\sum_{c=1}^{K} Pr(y=c|x) = 1$$

$$Pr(y=c|x) \geq 0 \quad \forall c$$

$$\exp\left[\log\left[\frac{Pr(y=c|x)}{Pr(y=K|x)}\right]\right] = \exp\left[w_c^T x + w_{c0}\right]$$

$$\log\left[\frac{p(x|y=c)}{p(x|y=K)}\right] + \log\left[\frac{Pr(y=c)}{Pr(y=K)}\right]$$

$$Pr(y=1|x) + Pr(y=2|x) + \dots + Pr(y=K-1|x) + Pr(y=K|x) = 1$$

$$\frac{Pr(y=1|x) + Pr(y=2|x) + \dots + Pr(y=K-1|x)}{Pr(y=K|x)} = \frac{1 - Pr(y=K|x)}{Pr(y=K|x)}$$

$$\sum_{c=1}^{K-1} \frac{Pr(y=c|x)}{Pr(y=K|x)} = \frac{1}{Pr(y=K|x)} - 1$$

$$\frac{Pr(y=c|x)}{Pr(y=K|x)} = \exp\left[w_c^T x + w_{c0}\right]$$

$$Pr(y=K|x) = \frac{1}{1+\sum_{c=1}^{K-1} \exp[w_c^T x + w_{c0}]}$$

$$Pr(y=c|x) = \frac{\exp[w_c^T . x + w_{c0}]}{1+\sum_{d=1}^{K-1} \exp[w_d^T . x + w_{d0}]} \quad c=1,2,..,K-1$$

$\theta = \{ \underbrace{w_1}_{D\times 1}, \underbrace{w_{10}}_{1\times 1}, \underbrace{w_2}_{D\times 1}, \underbrace{w_{20}}_{1\times 1}, \dots, \underbrace{w_{(K-1)}}_{D\times 1}, \underbrace{w_{(K-1)0}}_{1\times 1} \}$ → total # of parameters $= (K-1)(D+1)$

Assume $w_K^T . x + w_{K0} = 0$

Softmax $\left[Pr(y=c|x) = \frac{\exp(w_c^T . x + w_{c0})}{\sum_{d=1}^{K} \exp(w_d^T x + w_{d0})} \right] \forall c$

$$g_1(x_{N+1}) = w_1^T x_{N+1} + w_{10}$$
$$\vdots$$
$$g_K(x_{N+1}) = w_K^T x_{N+1} + w_{K0}$$

} pick the one with maximum score.

IDEAL SCORES

1 +2 $\hat{Pr}(y=1|x) = \frac{\exp(g_1(x))}{\exp(g_1(x))+\exp(g_2(x))+\exp(g_3(x))}$ 0.7214

0 -2 $\hat{Pr}(y=2|x) = \frac{\exp(g_2(x))}{\exp(g_1(x))+\exp(g_2(x))+\exp(g_3(x))}$ 0.0132

0 +1 $\hat{Pr}(y=3|x) = \frac{\exp(g_3(x))}{\exp(g_1(x))+\exp(g_2(x))+\exp(g_3(x))}$ 0.2654

+20

-20

+10

0.9999

0.0007

0.0003

+2000

-2000

+1000

$$\frac{\exp(2000)}{\exp(2000)+\exp(-2000)+\exp(1000)} = \frac{\infty}{\infty}$$

$\exp(2000) \to \infty$, $\exp(-2000) \to 0$, $\exp(1000) \to \infty$

$$\frac{1}{1+\exp[-score]}$$

it is always SAFE.

$$\frac{\exp(2000)\ \exp(-2000)}{[\exp(2000)+\exp(-2000)+\exp(1000)]\ \exp(-2000)}$$

$$\frac{\exp(0)}{\exp(0)+\exp(-4000)+\exp(-1000)} = \frac{1}{1} = 1$$

$\exp(0) \to 1$, $\exp(-4000) \to 0$, $\exp(-1000) \to 0$

$$y_i|x_i \sim \text{Multinomial}\left(y_i;\ 1,\ \{\Pr(y_i=c|x)\}_{c=1}^{K}\right)$$



$$\left(\frac{1}{6}\right)^{1}\left(\frac{1}{6}\right)^{0}\left(\frac{1}{6}\right)^{0}\left(\frac{1}{6}\right)^{0}\left(\frac{1}{6}\right)^{0}$$

$$\text{Multinomial}\left(y_i;\ 1,\ \{1/6, 1/6, 1/6, \ldots, 1/6\}\right)$$

$$\text{likelihood}\left(\{w_c, w_{c0}\}_{c=1}^{K} \mid \mathcal{X}\right) = \prod_{i=1}^{N}\prod_{c=1}^{K} \Pr(y_i=c|x)^{\mathbb{1}(y_i=c)}$$

$$\mathbb{1}(y_i=c) \equiv y_{ic}$$

$$= \prod_{i=1}^{N}\prod_{c=1}^{K} \Pr(y_i=c|x)^{y_{ic}}$$

$$y = \begin{bmatrix} 2 \\ 1 \\ 3 \\ 2 \end{bmatrix} \Rightarrow Y = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

one-hot encoding

$$\Downarrow$$

$$\hat{y}_{ic}$$

$$\log\left[\prod_{i=1}^{N}\prod_{c=1}^{K} \hat{y}_{ic}^{\,y_{ic}}\right]$$

$$\downarrow$$

$$\sum_{i=1}^{N}\sum_{c=1}^{K} y_{ic}\log(\hat{y}_{ic})$$

$$\text{loglikelihood} = \sum_{i=1}^{N}\sum_{c=1}^{K} y_{ic}\log\underbrace{\left[Pr(y_i=c|x_i)\right]}_{\hat{y}_{ic}}$$

$0.\log(0) \equiv 0$

$$\text{Error}(\{w_c, w_{c0}\}_{c=1}^{K} | \mathcal{X}) = -\sum_{i=1}^{N}\sum_{c=1}^{K} y_{ic}.\log(\hat{y}_{ic})$$

$$= -\sum_{i=1}^{N}\sum_{c=1}^{K} y_{ic}.\log\left[\frac{\exp\left[w_c^T.x_i + w_{c0}\right]}{\sum_{d=1}^{K}\exp\left[w_d^T.x_i + w_{d0}\right]}\right]$$

Exercise:

$$\frac{\partial \text{Error}}{\partial w_c} = ? \qquad \frac{\partial \text{Error}}{\partial w_{c0}} = ?$$

$$w_c^{(t+1)} = w_c^{(t)} + \Delta w_c$$

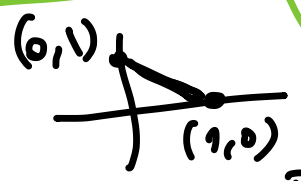
$$w_{c0}^{(t+1)} = w_{c0}^{(t)} + \Delta w_{c0}$$

$$\Delta w_d = \eta \sum_{i=1}^{N} \sum_{c=1}^{K} \frac{y_{ic}}{\sum_c y_{ic}} \cdot \hat{y}_{ic} \left[1(c=d) - \hat{y}_{id} \right] \cdot x_i$$

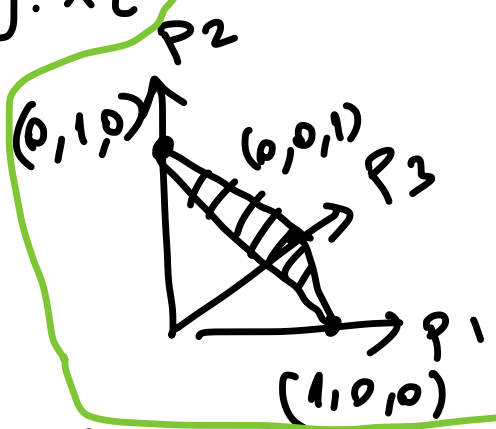
$$= \eta \sum_{i=1}^{N} \left[y_{id} - \hat{y}_{id} \right] \cdot x_i$$

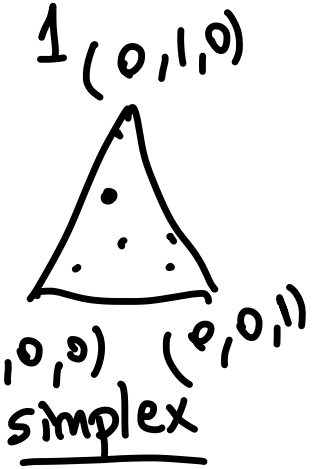
$$\Delta w_{d0} = \eta \sum_{i=1}^{N} \left[y_{id} - \hat{y}_{id} \right]$$

$p_1 + p_2 = 1$
$p_1 \geq 0$
$p_2 \geq 0$



$p_1 + p_2 + p_3 = 1$
$p_1 \geq 0$
$p_2 \geq 0$
$p_3 \geq 0$





simplex

ALGORITHM:

STEP #1: Initialize $\{w_1, w_{10}, w_2, w_{20}, \ldots, w_K, w_{K0}\}$ randomly to small values.

STEP #2: Calculate gradients

STEP #3: Update parameters using gradients

STEP #4: Go to STEP #2 if there is enough change in the parameters.

Linear Discrimination

Classification $\Rightarrow$ $\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$

$x_i \in \mathbb{R}$ → univariate

$x_i \in \mathbb{R}^D$ → multivariate

$y_i \in \{1, 2, \dots, K\}$

$g_1(x)$
$g_2(x)$
$\vdots$
$g_K(x)$
} score functions

$$\hat{y}_{N+1} = \arg\max_{c=1}^{K} g_c(x_{N+1})$$

$$g_c(x) = p(x|y=c)\,Pr(y=c)$$

$Pr(y=c) = \frac{N_c}{N}$

N_c → # of data points in Class c

N → total # of data points

$p(x|y=c)$ → multivariate ($x_i \in \mathbb{R}^D$)

$p(x|y=c)$ → univariate ($x_i \in \mathbb{R}$)

$N(x_i; \mu_c, \sigma_c^2)$

$\hat{\mu}_c, \hat{\sigma}_c^2$ (Lecture 3)

$N(x_i; \mu_c, \Sigma_c)$

$\hat{\mu}_c, \hat{\Sigma}_c$ (Lecture 5)

$$g_c(x \mid W_c, w_c, w_{c0}) = x^T . \underbrace{W_c}_{D \times D} . x + \underbrace{w_c^T}_{D \times 1} . x + \underbrace{w_{c0}}_{1 \times 1}$$

total # of model parameters $= K\left[\frac{D(D+1)}{2} + D + 1\right]$

$a^2 + 2ab + b^2$

$$\begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 1 & -5 \\ 7 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} a-5b \\ 7a+b \end{bmatrix}$$

$a^2 - 5ab + 7ab + b^2$

$a^2 + 2ab + b^2$

$$g_c(x \mid w_c, w_{c0}) = w_c^T . x + w_{c0}$$

total # of model parameters $= K[D+1]$

$$= \begin{bmatrix} w_{c1} & w_{c2} & \cdots & w_{cD} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_D \end{bmatrix} + w_{c0}$$

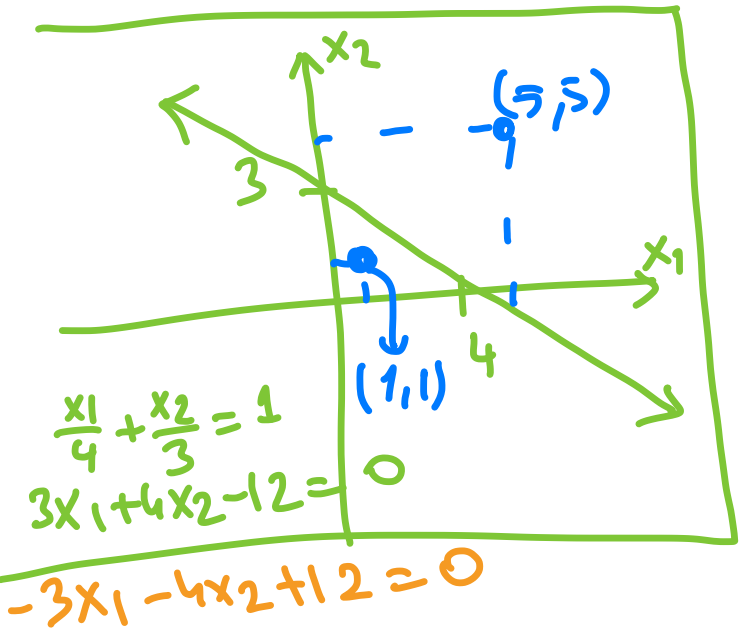
$$= \sum_{d=1}^{D} [w_{cd} . x_d] + w_{c0}$$

Binary Classification (K = 2)

$g_1(x)$, $g_2(x)$

if $g_1(x) > g_2(x) \Rightarrow \hat{y} = 1$

if $g_2(x) > g_1(x) \Rightarrow \hat{y} = 2$



if $g_1(x) - g_2(x) > 0 \Rightarrow \hat{y} = 1$

if $g_1(x) - g_2(x) < 0 \Rightarrow \hat{y} = 2$

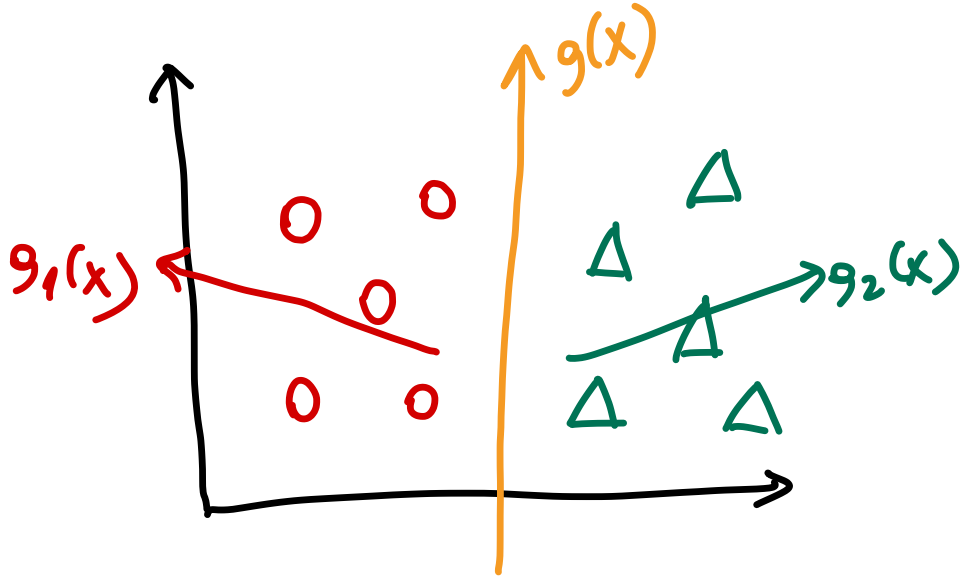
if $g(x) > 0 \Rightarrow \hat{y} = 1$

if $g(x) < 0 \Rightarrow \hat{y} = 2$

$g_1(x) = w_1^T . x + w_{10}$

$g_2(x) = w_2^T . x + w_{20}$

$g_1(x) - g_2(x) = (w_1 - w_2)^T . x + (w_{10} - w_{20})$



Multiclass classification ($K > 2$)

$g_1(x)$
$g_2(x)$
$\vdots$
$g_K(x)$

$\hat{y} = \arg\max_{c=1}^{K} g_c(x)$

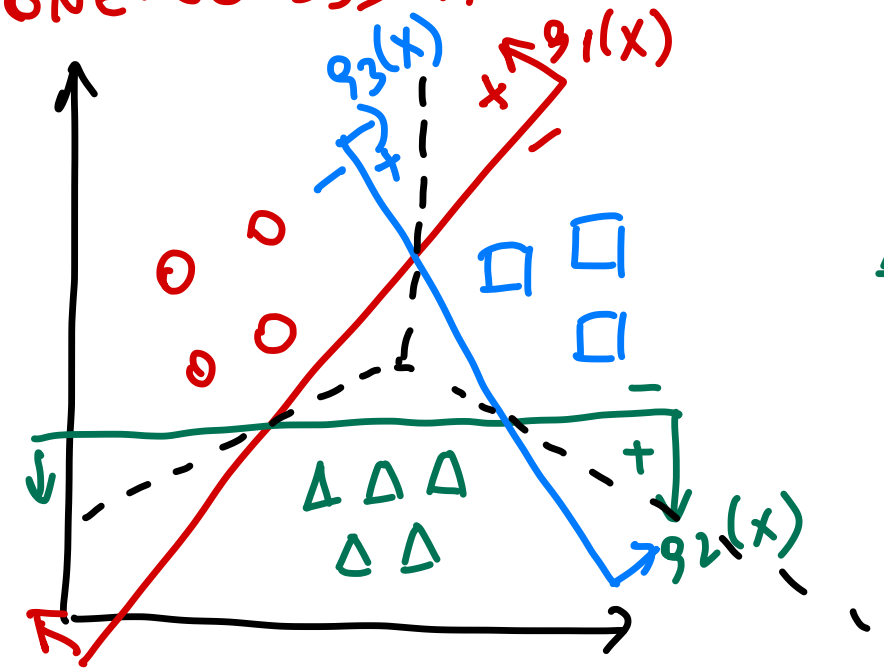
assume $K=3$

if $g_1(x) > g_2(x)$ & $g_1(x) > g_3(x)$ $\Rightarrow \hat{y} = 1$

if $g_2(x) > g_1(x)$ & $g_2(x) > g_3(x)$ $\Rightarrow \hat{y} = 2$

if $g_3(x) > g_1(x)$ & $g_3(x) > g_2(x)$ $\Rightarrow \hat{y} = 3$

ONE-VERSUS-ALL (OVA) APPROACH



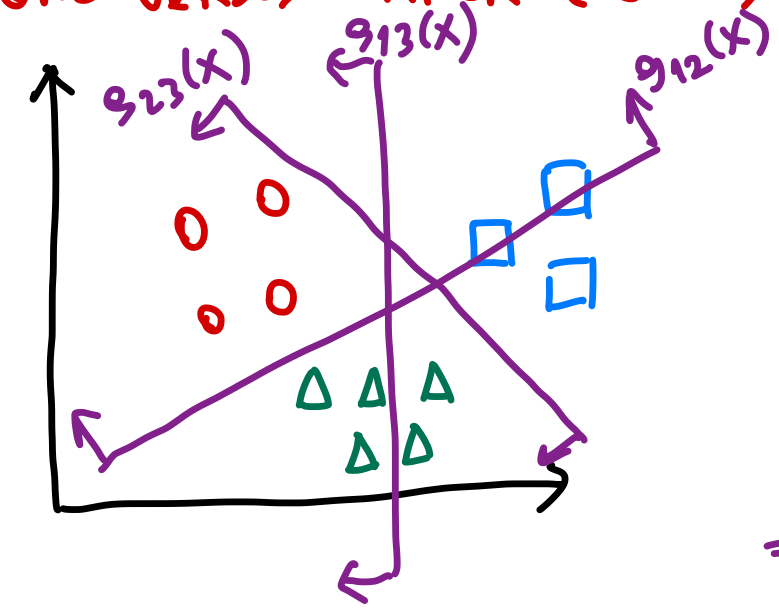
+	−
OOOO	ΔΔΔΔΔDDD $\Rightarrow N = 12$
ΔΔΔΔΔ	OOOO DDD $\Rightarrow N = 12$
DDD	OOOO ΔΔΔΔΔ $\Rightarrow N = 12$

of model parameters $= K(D+1)$

data set size $= N$

of score functions $= K$

ONE-VERSUS-OTHER (OVO) APPROACH



	+	−	
$g_{12}(x)$	OOOO	ΔΔΔΔΔ	$\Rightarrow N = 9$
$g_{13}(x)$	OOOO	□□□	$\Rightarrow N = 7$
$g_{23}(x)$	ΔΔΔΔΔ	□□□	$\Rightarrow N = 8$

$$\#\text{ of model parameters} = \frac{K(K-1)}{2}(D+1)$$

$$\text{data set size} = 2 \cdot \frac{N}{K}$$

$$\#\text{of score functions} = \frac{K(K-1)}{2}$$

X	1	2	3
$g_{12}(x)$	1	0	∅
$g_{13}(x)$	1	∅	0
$g_{23}(x)$	∅	0	1
# of wins $\Rightarrow$	2	0	1

→ pick the one with the highest # of wins.

1 : win

0 : loss

∅ : no participation

K=2

$$Pr(y=1|x) = \delta$$

$$Pr(y=2|x) = 1-\delta$$

$$\log\left(\frac{a\cdot b}{c\cdot d}\right) = \log\left(\frac{a}{c}\right) + \log\left(\frac{b}{d}\right)$$

if $\delta > 0.5 \Rightarrow \hat{y} = 1$

odds ratio ← if $\frac{\delta}{1-\delta} > 1 \Rightarrow \hat{y} = 1$

log-odds ratio ← if $\log\left[\frac{\delta}{1-\delta}\right] > 0 \Rightarrow \hat{y} = 1$

$$\log\left[\frac{Pr(y=1|x)}{Pr(y=2|x)}\right] = \log\left[\frac{p(x|y=1)\,Pr(y=1)\,/p(x)}{p(x|y=2)\,Pr(y=2)\,/p(x)}\right]$$

$$N(x;\mu_1,\Sigma_1) = \log\left[\frac{p(x|y=1)}{p(x|y=2)}\right] + \log\left[\frac{Pr(y=1)}{Pr(y=2)}\right]$$

$$N(x;\mu_2,\Sigma_2)$$

Assuming $\Sigma_1 = \Sigma_2 = \Sigma$ (shared covariance assumption)

$$p(x;\mu,\Sigma) = \frac{1}{\sqrt{(2\pi)^p|\Sigma|}} \cdot \exp\left[-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right]$$

$$= \log\left[\frac{\frac{1}{\sqrt{(2\pi)^p|\Sigma|}}\exp\left[-\frac{1}{2}(x-\mu_1)^T\Sigma^{-1}(x-\mu_1)\right]}{\frac{1}{\sqrt{(2\pi)^p|\Sigma|}}\exp\left[-\frac{1}{2}(x-\mu_2)^T\Sigma^{-1}(x-\mu_2)\right]}\right] + \log\left[\frac{Pr(y=1)}{Pr(y=2)}\right]$$

$$= -\frac{1}{2}(x-\mu_1)^T\Sigma^{-1}(x-\mu_1) + \frac{1}{2}(x-\mu_2)^T\Sigma^{-1}(x-\mu_2) + \log\left[\frac{Pr(y=1)}{Pr(y=2)}\right]$$

$$= -\frac{1}{2}x^T\Sigma^{-1}\cdot x + \mu_1^T\Sigma^{-1}\cdot x - \frac{1}{2}\mu_1^T\Sigma^{-1}\mu_1 + \frac{1}{2}x^T\Sigma^{-1}\cdot x - \mu_2^T\cdot\Sigma^{-1}\cdot x - \frac{1}{2}\mu_2^T\Sigma^{-1}\cdot\mu_2 + \dots$$

linear | constant | linear | constant + constant

$$(\mu_1-\mu_2)^T\cdot\Sigma^{-1}\cdot x = w^T$$

$$(A\cdot B)^T = B^T A^T$$

$$= W^T\cdot X + W_0$$

$$\hat{w} = \hat{\Sigma}^{-1}\cdot(\hat{\mu}_1-\hat{\mu}_2)$$

$$\hat{w}_0 = -\frac{1}{2}(\hat{\mu}_1+\hat{\mu}_2)^T\hat{\Sigma}^{-1}(\hat{\mu}_1-\hat{\mu}_2) + \log\left[\frac{\hat{Pr}(y=1)}{\hat{Pr}(y=2)}\right]$$

Linear Discrimination

$$g(x) = w^T \cdot x + w_0$$

by assuming $p(x|y=1) = N(x; \mu_1, \Sigma)$, $p(x|y=2) = N(x; \mu_2, \Sigma)$

$\hat{\mu}_1$ → sample mean of class1

$$\underset{D\times 1}{\hat{w}} = \hat{\Sigma}^{-1} \cdot (\hat{\mu}_1 - \hat{\mu}_2)$$

$\hat{\mu}_2$ → sample mean of class2

$$\underset{1\times 1}{\hat{w}_0} = -\frac{1}{2}(\hat{\mu}_1 + \hat{\mu}_2)\hat{\Sigma}^{-1}(\hat{\mu}_1 - \hat{\mu}_2) + \log\left[\frac{\hat{Pr}(y=1)}{\hat{Pr}(y=2)}\right]$$

$\hat{\Sigma}$ → sample covariance of all data points

$\hat{Pr}(y=1)$ → frequency of class 1

$\hat{Pr}(y=2)$ → frequency of class 2

$$\exp\left[\log\left[\frac{\delta}{1-\delta}\right]\right] = \exp[w^T \cdot x + w_0]$$

$$\frac{\delta}{1-\delta} = \exp[w^T \cdot x + w_0] \Rightarrow \delta = \exp[w^T \cdot x + w_0] - \delta\exp[w^T \cdot x + w_0]$$

$$\delta + \delta\exp[w^T \cdot x + w_0] = \exp[w^T \cdot x + w_0]$$

$$\delta(1 + \exp[w^T \cdot x + w_0]) = \exp[w^T \cdot x + w_0]$$

$$\delta = \frac{\exp[w^T \cdot x + w_0]\exp[-(w^T \cdot x + w_0)]}{1 + \exp[w^T \cdot x + w_0]\exp[-(w^T \cdot x + w_0)]}$$

$$\delta = \frac{1}{1 + \exp[-(w^T \cdot x + w_0)]}$$

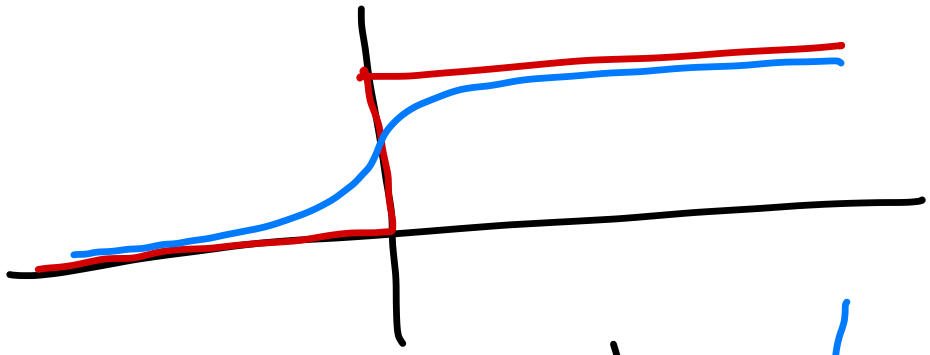
Sigmoid

i) $w^T.x + w_0 > 0 \Rightarrow \delta > 0.5$ $\quad 1-\delta < 0.5$

ii) $w^T.x + w_0 = 0 \Rightarrow \delta = 0.5$ $\quad 1-\delta = 0.5$

iii) $w^T.x + w_0 < 0 \Rightarrow \delta < 0.5$ $\quad 1-\delta > 0.5$

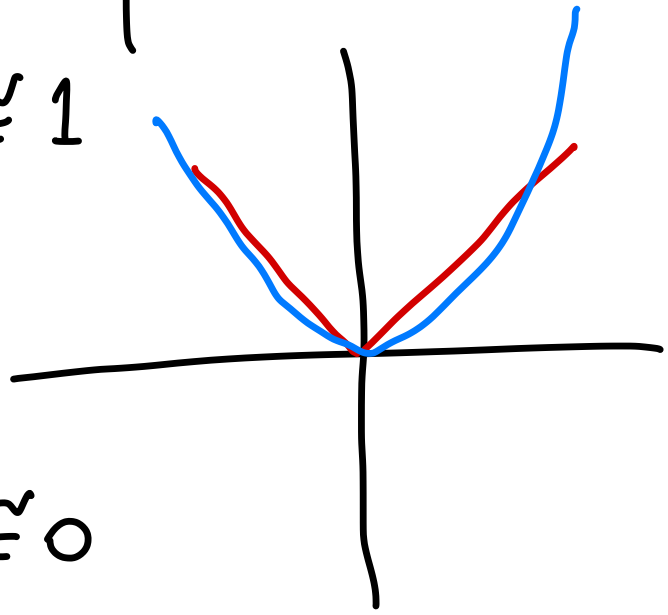
$$\delta(a) = \frac{1}{1+\exp[-a]}$$



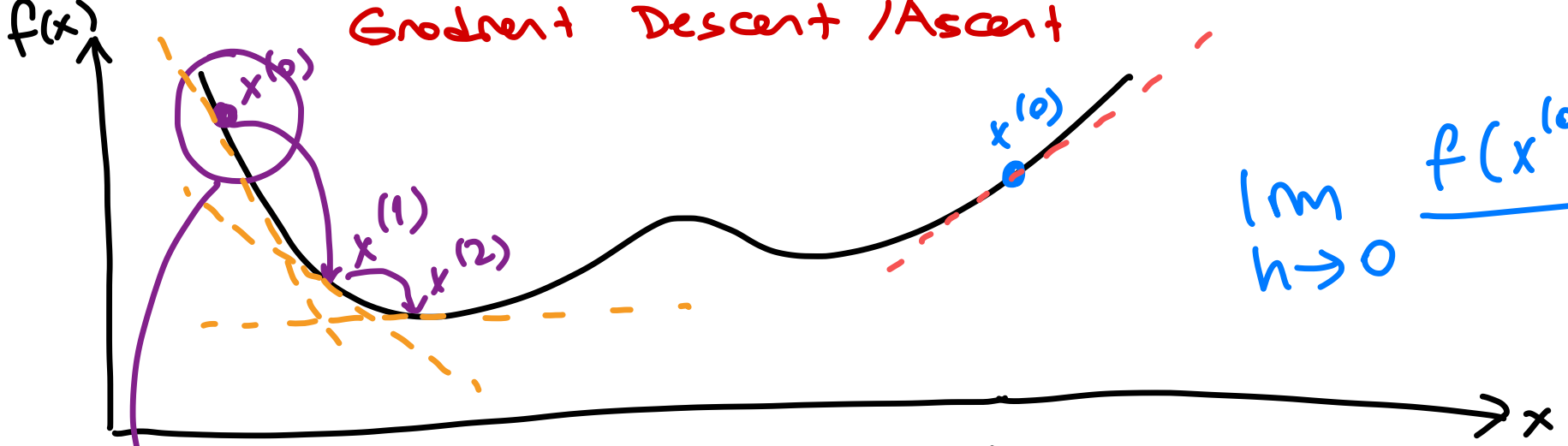
if $a=+5 \Rightarrow \frac{1}{1+\exp(-5)} \simeq 0.9933 \simeq 1$

if $a=0 \Rightarrow \frac{1}{1+\exp(0)} = 0.5$

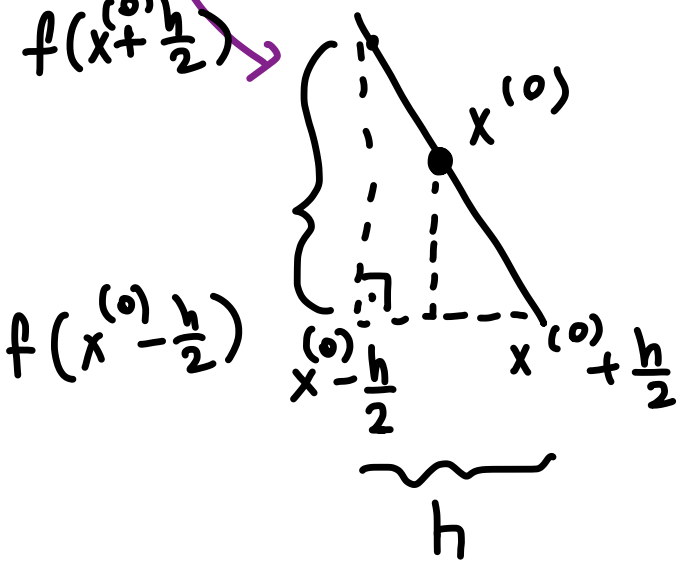
if $a=-5 \Rightarrow \frac{1}{1+\exp(+5)} \simeq 0.0067 \simeq 0$



Gradient Descent / Ascent



$$\lim_{h \to 0} \frac{f(x^{(0)}+h) - f(x^{(0)})}{h}$$



$$\lim_{h \to 0} \frac{f(x^{(0)}+\frac{h}{2}) - f(x^{(0)}-\frac{h}{2})}{h}$$

stepsize

$$\Delta x = -\eta \cdot \frac{\partial f(x)}{\partial x}$$

$$x^{(t+1)} = x^{(t)} - \eta \cdot \frac{\partial f(x)}{\partial x}$$

update equation

$$(w^*, w_0^*) = \underset{(w,w_0)}{\arg\min}\ E[w, w_0 \mid \mathcal{X}]$$

↳ training set

$$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N} \qquad x_i \in \mathbb{R}^D \qquad y_i \in \{0, 1\}$$

failure ← 0, 1 → success

$$y_i | x_i \sim \text{Bernoulli}\left(y_i;\ \hat{Pr}(y_i = 1 \mid x_i)\right)$$

$\hat{y}_i$ → predicted posterior probability.

$$\text{Likelihood}(w, w_0 \mid \mathcal{X}) = \prod_{i=1}^{N} \left[\hat{y}_i^{\,y_i} \cdot (1 - \hat{y}_i)^{1 - y_i}\right]$$

$$\log \text{likelihood}(w, w_0 \mid \mathcal{X}) = \sum_{i=1}^{N} \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)\right]$$

$$E[w, w_0 \mid \mathcal{X}) = -\sum_{i=1}^{N} \left[\underbrace{y_i}_{\text{constant}} \underbrace{\log(\hat{y}_i)}_{f(w,w_0)} + \underbrace{(1 - y_i)}_{\text{constant}} \underbrace{\log(1 - \hat{y}_i)}_{f(w,w_0)}\right]$$

negative log likelihood / error / loss

minimize $-\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right]$

with respect to w, w_0

$$\hat{y}_i = \frac{1}{1+\exp\left[-(w^T x_i + w_0)\right]}$$

$$\frac{\partial E[w, w_0 | \mathcal{X}]}{\partial w} = ? \qquad \frac{\partial E[w, w_0 | \mathcal{X}]}{\partial w_0} = ?$$

$$\frac{\partial \log(\hat{y}_i)}{\partial w} = \frac{\partial \log(\hat{y}_i)}{\partial \hat{y}_i} \frac{\partial \hat{y}_i}{\partial w}$$

$$= \frac{1}{\hat{y}_i} \cdot \hat{y}_i (1-\hat{y}_i) \cdot x_i$$

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$$= (1-\hat{y}_i) \cdot x_i$$

$$\frac{\partial \log(a)}{\partial a} = \frac{1}{a}$$

$$\frac{\partial \frac{f(a)}{g(a)}}{\partial a} = \frac{\frac{\partial f(a)}{\partial a} g(a) - \frac{\partial g(a)}{\partial a} f(a)}{[g(a)]^2}$$

$$\text{sigmoid}(a) = \frac{1}{1+\exp(-a)}$$

Show that $\frac{\partial\, \text{sigmoid}(a)}{\partial (a)} = \text{sigmoid}(a)\left[1 - \text{sigmoid}(a)\right]$

$$\frac{\partial\, \text{sigmoid}(a)}{\partial\, a} = \frac{\frac{\partial\, 1}{\partial\, a}\cdot\left[1+\exp(-a)\right] - \frac{\partial\left(1+\exp(-a)\right)}{\partial\, a}\cdot 1}{\left[1+\exp(-a)\right]^2}$$

($\frac{\partial\, 1}{\partial\, a} = 0$)

$w^T \cdot x = x^T \cdot w$

$$= \frac{1\,(-1)\exp(-a)}{\left[1+\exp(-a)\right]^2} = \underbrace{\frac{1}{1+\exp(-a)}}_{\text{sigmoid}(a)} \cdot \underbrace{\frac{\exp(-a)}{1+\exp(-a)}}_{\left[1-\text{sigmoid}(a)\right]}$$

$$\frac{\partial\, \text{sigmoid}(\overbrace{w^T \cdot x_i + w_0}^{a})}{\partial\, w} = \frac{\partial\, \text{sigmoid}(a)}{\partial\, a}\,\frac{\partial\, a}{\partial\, w}$$

$$= \underbrace{\text{sigmoid}(w^T \cdot x_i + w_0)}_{\hat{y}_i}\cdot\left[\underbrace{1 - \text{sigmoid}(w^T \cdot x_i + w_0)}_{(1-\hat{y}_i)}\right]\cdot \underbrace{x_i}_{x_i}$$

Exercise $\frac{\partial \log(\hat{y}_i)}{\partial w_0} = (1-\hat{y}_i)$

Exercise $\frac{\partial \log(1-\hat{y}_i)}{\partial w} = ?$ $\frac{\partial \log(1-\hat{y}_i)}{\partial w_0} = ?$

$$\frac{\partial Error[w, w_0 | \mathcal{X}]}{\partial w} = -\sum_{i=1}^{N} \left[y_i (1-\hat{y}_i) \cdot x_i + (1-y_i) \boxed{\quad} \right]$$

$$= \boxed{-\sum_{i=1}^{N} (y_i - \hat{y}_i) \cdot x_i}$$

$$\frac{\partial Error[w, w_0 | \mathcal{X}]}{\partial w_0} = \boxed{-\sum_{i=1}^{N} (y_i - \hat{y}_i)}$$

$$w^{(t+1)} = w^{(t)} - \eta \cdot \frac{\partial Error}{\partial w}$$

$$w_0^{(t+1)} = w_0^{(t)} - \eta \cdot \frac{\partial Error}{\partial w_0}$$

